Unit: Unit 9: Data Statistical Analysis
Topic: Measures of Central Tendency

Name: ________________
Date: ___________

Advanced (Habanero)

Q1. What is the mean of $2, 4, 6$?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q2. Which measure of central tendency is most affected by outliers?
(A) Mean
(B) Median
(C) Mode
(D) Range
(E) Variance

Q3. A dataset has $5, 5, 5, 10$; what is the mode?
(A) 5
(B) 10
(C) 7.5
(D) 5.5
(E) No mode

Q4. What is the median of $1, 3, 5, 7, 9$?
(A) 1
(B) 3
(C) 5
(D) 7
(E) 9

Q5. A set of exam scores has a mean of 80 and a median of 75; what can be inferred?
(A) Scores are normally distributed
(B) Scores are skewed right
(C) Scores are skewed left
(D) No inference can be made
(E) Mean is less than median

Q6. Which of the following is a characteristic of the mode?
(A) It is always unique
(B) It is always the mean
(C) A dataset may have more than one mode
(D) It is the most frequently occurring value
(E) It is the middle value

Q7. What is the purpose of using measures of central tendency?
(A) To describe the spread of the data
(B) To identify the most frequent value
(C) To summarize the data with a single value
(D) To determine the median
(E) To create a box plot

Q8. A dataset has the following values: $10, 12, 15, 18$; what is the mean?
(A) 11
(B) 12
(C) 13.5
(D) 13.75
(E) 14

Open Ended Questions (FRQ)

Q9. Determine the mean, median, and mode of the dataset $2, 4, 6, 8, 10, 12$ and describe what each measure indicates about the data.

Q10. Compare the effectiveness of the mean, median, and mode in describing the central tendency of a skewed distribution and a symmetric distribution; provide examples.

Chef's Tips

- Tip 1: Ensure students understand the definitions and differences between mean, median, and mode
- Tip 2: Use real-world examples to illustrate the application of measures of central tendency
- Tip 3: Encourage students to consider the context when selecting an appropriate measure of central tendency